

Week 2 - BFSI & AI: Use-Case & Solution-Flow

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Payments & Digital Lending

Part A: Pain-Point Map

Pain Point	Impact	Root Cause (5 Whys)	AI-Solvable?	Use Case Type
High False Decline Rate	Revenue loss & customer churn.	Rigid, static rules lack context.	Yes	Anomaly Detection.
Manual Fraud Investigation	High operational costs.	Inefficient manual data triage.	Yes	Classification (Scoring).
Data Siloing	Incomplete risk visibility.	Disparate legacy architecture.	No	Integration Problem.

High False Decline Rate:

- Instead of blocking transactions based on rigid, binary rules like 'Block all transactions > ₹50,000', AI uses Gradient Boosted Trees (GBT) to analyze hundreds of features in real-time.

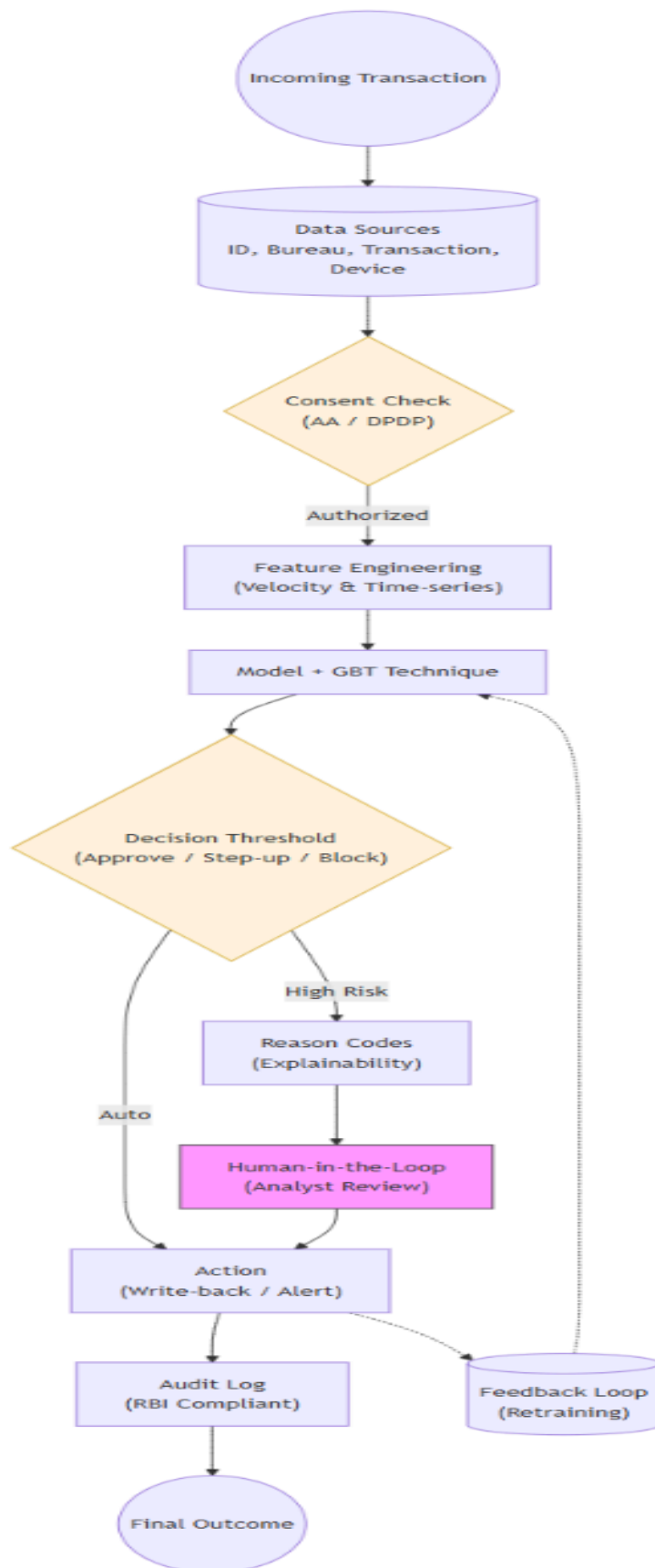
Manual Fraud Investigation:

- AI models assign a risk score to every transaction.

Inefficient Data Triaging:

- AI pipelines automate the ingestion and feature engineering of unstructured data from disparate sources like device telemetry, transaction history, and geo-location into a single, usable dataset.

Part B: Solution Flowchart



1. **Trigger:** Incoming transaction (e.g., UPI).
2. **Data Sources:** Transaction stream, device metadata, geo-location, historical behavior.
3. **Consent Check:** DPDP/Account Aggregator check performed before ingestion.
4. **Feature Engineering:** Raw data transformed into velocity and time-series features.
5. **Model:** Anomaly Detection + Gradient Boosted Trees (GBT).
6. **Decision Point:** Confidence threshold applied (Auto-approve vs. Step-up vs. Block).
7. **Explainability:** Reason codes generated to justify decisions.
8. **Human-in-the-loop:** Analyst reviews high-risk flagged exceptions.
9. **Audit Log:** Every automated decision recorded for RBI compliance.
10. **Feedback Loop:** Confirmed fraud/repayment outcomes used to retrain the model.

Part C: Justification Note

- **Why this technique?**
Gradient Boosted Trees are highly effective for tabular transaction data and allow for the generation of reason codes via SHAP or similar methods, which are mandatory for regulatory explainability.
- **The Error Dial:**
Decisions are calibrated to minimize False Negatives (missed fraud) while accounting for the cost of False Positives (customer friction). Decisions are based on financial impact in rupees, rather than generic accuracy.
- **Data & Consent:**
Data is sourced from transaction streams and device telemetry. We adhere to the Account Aggregator (AA) framework, ensuring all data usage is consented and time-bound.

Fairness & Top 3 Risks:

1. **Proxy Bias:** Risk that features (e.g., device type) cause disparate impact. *Mitigation:* Regular testing for bias across customer segments.
2. **Model Drift:** Rapid changes in fraud patterns. *Mitigation:* Daily monitoring of precision/recall metrics.
3. **Customer Friction:** Unnecessary step-ups for legitimate users.

MVP: The MVP will be deployed on 10% of traffic, focusing initially on low-value UPI transactions, with a mandatory human audit loop for every blocked transaction